

CODE ADDITIONS

7/8/2026; 7:53 pm (Isaac)

- Implemented an image processing algorithm that (mostly) works for Tube images. It needs testing.

7/9/2026; 4:34 pm (Ryan)

- Updated Dataset organization so that each subfolders organization matches. Changed to imageDatastore and added class labels and pass/fail classification for each image.

7/11/2026; 4:14 pm (Ryan)

- Added corner removal to improve output decision reliability. Added comments to better explain which image is which when selected from the range (Tubes only).

7/12/2026; 10:52 am (Ryan)

- Added labels to ROI that correspond to each table entry for easier analysis of performance. Adjusted minimum eccentricity to improve detection of short tubes as anomalous.

7/16/2026; 4:06 pm (Ryan)

- Adjusted the singleimageinspectionfunction to run without displaying output, allowing for faster results to be gathered for batch processing. Basic batch processing implemented, will need further improvements. Corrected some program errors.

7/19/2026; 11:04 am (Ryan)

- Added yield and defect rates and also a pass/fail confusion matrix. Separated batch inspection into its own function and removed unnecessary code.

Choose Data Subset of Interest, Adjust Parameters

```
dataSubset = "tubes";  
displayOriginalImage =true;  
useAIClassifier =true;  
  
performBatchTesting =true;
```

Current working algorithm

```
% Obtain the file location of the dataset.  
% ***IMPORTANT***  
% This file location stored is a RELATIVE path!!!  
% For this script to run, your active path must be set to where  
% the MPDD dataset is stored on your local system.  
%  
% To change your active path, click the "Set Path" option under the  
% "Environment" sub tab within the MATLAB application.
```

```

fileLoc = sprintf("data/%s", dataSubset);

% Creates an image Datastore to store all images of specified subset type.
% The image Datastore is setup to specifically look for images with certain file
% extensions that are common for images (i.e. jpg, png, etc.)
ds = imageDatastore(fileLoc, 'IncludeSubfolders',true, 'LabelSource','foldernames',
...
    'FileExtensions', [".jpg", ".png", ".tiff"]);

% Filter 'ground truth'/masks that are only used for model verification
excludeMask = contains(ds.Files, "gt*" | "mask");
ds = subset(ds, ~excludeMask);

% Obtain the number of files within the selected sub directory
numFiles = numel(ds.Files);

% ***Image processing begins here***
%
% Load the image of interest from the datastore object
% Color mismatch      01 - 008, + 21
% Defective Length    09 - 021
% Malformed Metal     54 - 103, + 8
% Good                22 - 053, + 104 - 225
% NOTE: changes made to folders have made these ranges unreliable
imgNum = 20;
img = readimage(ds, imgNum);

% Labels image and (Optionally) Displays class & image info
% imgClass <-- Official label of the image
% imgPF <-- Whether or not the part would pass or fail
imgClass = ds.Labels(imgNum);

if displayOriginalImage
    numObsPerClass = countEachLabel(ds)
    imgInfo = sprintf("Image class: %s", imgClass);

    disp("Number of Files: " + numFiles);
    imshow(img);
    title(imgInfo);
end

% Inspect the image and print out a report
reportData = singleImageInspectionFunction(img, useAIClassifier);
displayInspectionOutput(img, reportData);

% Batch image classifier
%if performBatchTesting
%    runBatchInspection(ds);
%end

```